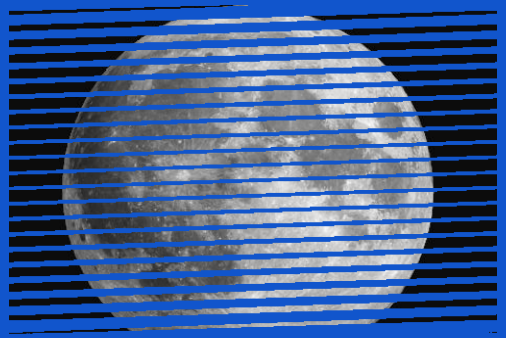
Space Exploration



Artemis II

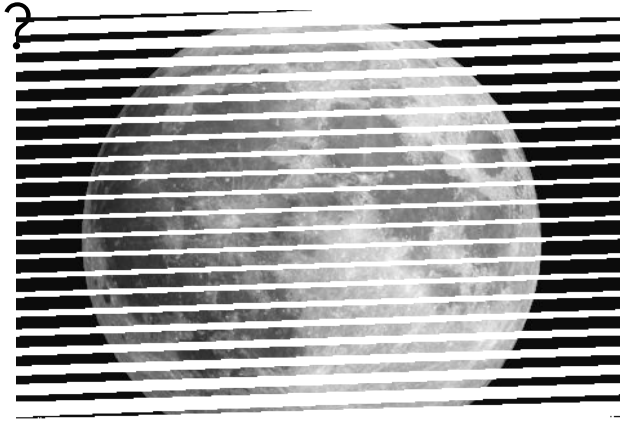
Why do we explore space?

ARE WE THE
ONLY SPECIES
IN THE COSMOS

Our missions to the moon…

Discussion questions:

1) When did we first go to the moon?
2) How many times have we been to the moon?
3) Why did we decide to go to the moon?



WHY THE MOON

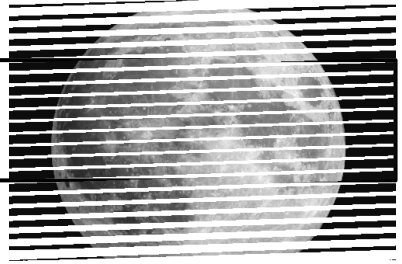
Our missions to the moon... a brief history

Before we look forward to Artemis II and our return to the Moon, we have to look back at the giants whose shoulders we're standing on.

The **Apollo Program** wasn't just a series of space flights; it was a massive leap in engineering, physics, and human endurance.

How did we first make it to the moon?

Our missions to the moon... a brief history



The "Space Race"

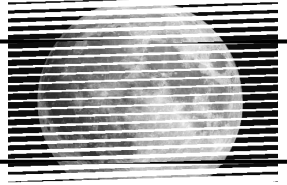
In the **1960s**, the **U.S.** and the **Soviet Union** were in a heated competition. In 1961, President John F. Kennedy set a bold goal: **landing a man on the Moon and returning him safely to Earth** before the decade was out. At the time, we barely knew if humans could survive in space for more than a few hours!

To get out of Earth's gravity, we needed a beast of a machine. The **Saturn V** remains one of the most powerful rockets ever built.

- Height: Over **110 meters** (363 feet) tall.
- Stages: It used a **three-stage** system where parts of the rocket fell away once their fuel was spent to **shed weight**.
- Fuel: It burned massive amounts of liquid **hydrogen** and liquid **oxygen**.

Saturn V

Our missions to the moon... a brief history



Apollos 1 through 17 - While there were 17 numbered missions, there were a few that had very significant impacts...

Apollo 1 (Jan 27, 1967)	A tragic fire during a ground test killed three astronauts, leading to massive safety redesigns.
Apollo 8 (Dec 21, 1968)	The first time humans left Earth's orbit, circled the Moon, and saw an "Earthrise."
Apollo 11 (Jul 16, 1969)	Neil Armstrong and Buzz Aldrin become the first humans to walk on the Moon.
Apollo 13 (Apr, 1970)	The "Successful Failure." An oxygen tank exploded, but the crew used the Lunar Module as a lifeboat to return home.
Apollo 17 (Dec 7, 1972)	The final mission. Gene Cernan was the last human to leave a footprint on the lunar surface... until now.

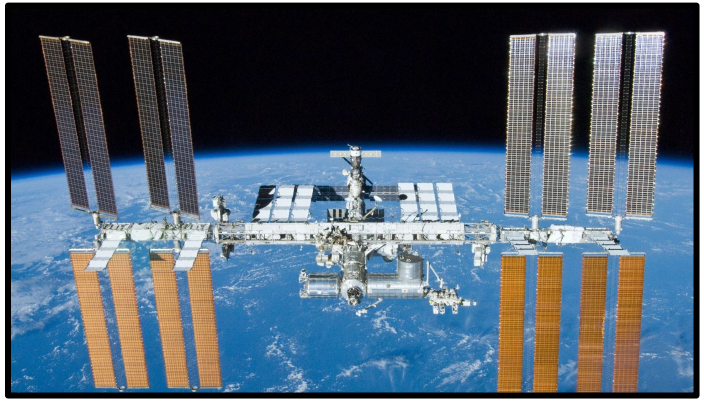
Our missions to the moon... a brief history

Why did we stop?

A common question is: *"If we went in the 60s, why haven't we been back?"*

It mostly came down to **cost and shifting priorities**. Once the "Race" was won, the massive funding (nearly 4% of the US federal budget at the time!) was scaled back. We shifted focus to the **International Space Station** and low-Earth orbit.

Notice the difference in technology. Your smartphone has millions of times more processing power than the computers that guided Apollo 11. That's why Artemis II is so exciting—we are going back with 21st-century tech!



Artemis II - Launched April 1st, 2026

Who's aboard?

Victor Glover
(*Pilot* & first person of color to travel to the moon)

Christina Koch
(*Mission Specialist* & first woman to travel to the moon)

Jeremy Hansen
(*Mission Specialist* & the first Canadian to travel to the moon)

Reid Wiseman
(*Commander* & veteran who previously lived on the ISS)

ARTEMIS II
RECAP

What was the Artemis II mission testing?

Artemis II conducted critical 10-day tests of the Orion spacecraft with a crew aboard to prepare for future lunar landings and preparing for future missions to Mars.

They will focus on :

- The Mission Context
- The Crew & Representation
- Technology & Engineering
- The Flight Path & Science
- Innovation & Sustainability
- Critical Thinking

Your Turn!

Mission Brief

You are part of a **post mission analysis team** reviewing the completed Artemis II mission. Your job is to investigate the mission using reliable sources and explain not just *what* happened but *why it matters for Earth and the future*.

You will work with your table partner to gather information about the Artemis II mission.